

9 (extra): James-Stein estimator

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Fact: $\chi^2(k)$ is the "chi-squared" distribution w/ k degrees-of-freedom ("df"), defined as the distribution of the sum of the squares of k iid $N(0,1)$ r.v.

Fact: If $Y \sim \chi^2(k)$ then $\frac{1}{Y}$ has the "inverse chi squared" or "inverted chi square" distribution w/ k d.f.
and its mean is $\frac{1}{k-2}$ if $k > 2$. cf. wikipedia

Stein's Paradox '55, James-Stein Estimator '61

Let $Y \sim N(\theta, \sigma^2 I_{m \times m}) \in \mathbb{R}^m$, $m \geq 3$. Goal: estimate θ from a realization y .

Obvious statistic is $\hat{\theta}_{LS} (= \hat{\theta}_{MLE}) := Y$. Unbiased.

But we can do better! $\hat{\theta}_{JS} := \left(1 - \frac{(m-2)\sigma^2}{\|Y\|^2}\right) Y$. Then $\exists \theta$ st.

$$\underbrace{\mathbb{E} \|\hat{\theta}_{JS} - \theta\|_2^2}_{\text{risk}} < \mathbb{E} \|\hat{\theta}_{LS} - \theta\|_2^2 \quad \dots \text{and hence LS is inadmissible.}$$

Counter-intuitive: bundling unrelated items can improve risk?!

One explanation: $\hat{\theta}_{JS}$ isn't better in all senses, it's better only for squared loss.

For any fixed metric, we can play mathematical tricks.

Quite possibly $\mathbb{E} \|\hat{\theta}_{JS} - \theta\|_1 > \mathbb{E} \|\hat{\theta}_{LS} - \theta\|_1$, for example.

In class exercise: for $m=3$, $\theta = \vec{0}$, $\sigma^2=1$, prove it! i.e. $\mathbb{E} \|\hat{\theta}_{JS}\|_2^2 < \mathbb{E} \|Y\|_2^2$

proof

$$\begin{aligned} \mathbb{E} \|\hat{\theta}_{JS}\|^2 &= \mathbb{E} \left\| Y - \frac{1}{\|Y\|^2} Y \right\|^2 = \mathbb{E} \left[\|Y\|^2 - 2 \frac{Y^T Y}{\|Y\|^2} + \frac{Y^T Y}{\|Y\|^4} \right] \\ &= \mathbb{E} \|Y\|^2 - 2 - \underbrace{\left(\mathbb{E} \frac{1}{\|Y\|^2} \right)}_{\text{inverse } \chi^2, \mathbb{E} \text{ is } \frac{1}{m-2} \text{ i.e. } \frac{1}{1} \text{ if } m=3} \\ &= \mathbb{E} \|Y\|^2 - 2 + 1 \quad \checkmark \end{aligned}$$

btw, $=3$. One way to see it,

$$\begin{aligned} \mathbb{E}[Y^T Y] &= \mathbb{E}[\text{tr}(Y^T Y)] \\ &= \mathbb{E}[\text{tr}(Y Y^T)] \quad \leftarrow \text{circular property of trace} \\ &= \text{tr}(\mathbb{E}[Y Y^T]) \\ &= \text{tr}(\sigma^2 I) \\ &= \sigma^2 \cdot m = 3 \end{aligned}$$